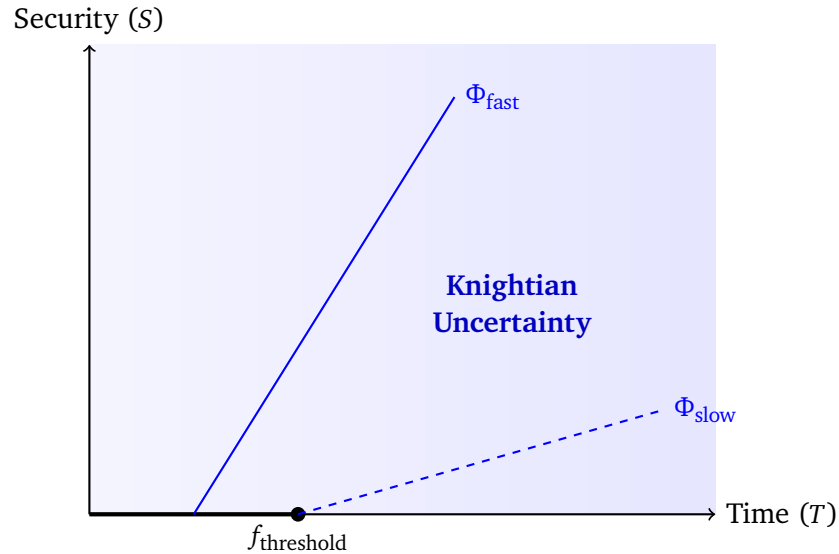


THE PROBLEM

Alice needs \$10,000 cost-of-revision in 24 hours. How much should she bid given that neither the inclusion time nor the rate at which security will accumulate can be known *ex ante*?



Alice has no truthful bid.

- others bid low? Alice prefers high.
- others bid high? Alice prefers low.

Nor is there a meaningful distribution of types.

- player “shadow bids” are always hidden.
- we cannot reconstruct distributions of types.

We have a game under Knightian Uncertainty.

- uncertainty over price aggregation;
- uncertainty over type distribution;
- uncertainty over the future;
- uncertainty over counterparty credibility (Hurwicz).

CAN WE ACHIEVE INCENTIVE COMPATIBILITY?

Yes. Implementation requires only two assumptions.

A. Monotonicity

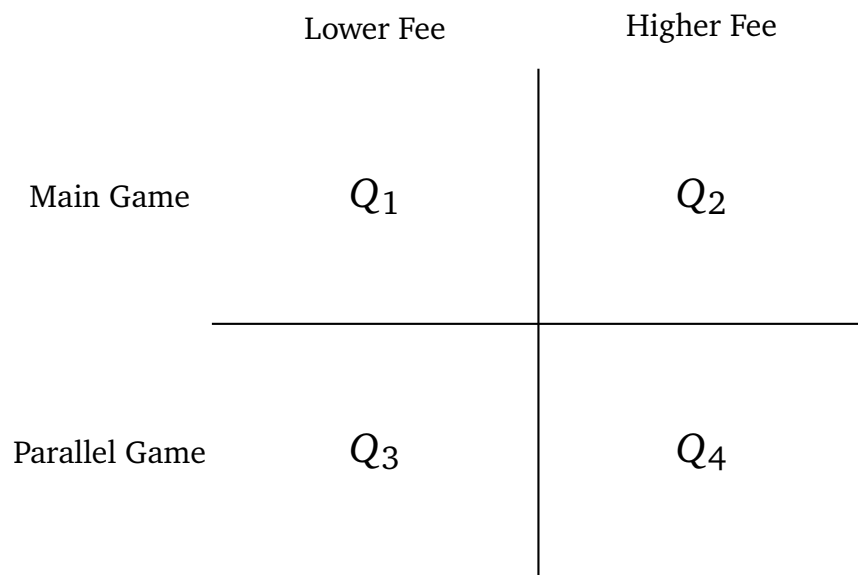
- Time Monotonicity: increasing the bid weakly increases expected speed of inclusion.
- Security Monotonicity: increasing the bid weakly increases the slope of the security curve.

B. Parallel Games:

- Main Game: bids are public, competition is higher, expected inclusion is faster.
- Parallel Game: bids are private, competition is lower, expected inclusion is slower.

Each player makes two decisions.

- which game to play.
- which bid to offer.



WHICH GAME TO PLAY?

Our two games differ in only one economically-relevant dimension.

The Main Game

- fastest inclusion in expectation;
- baseline for rational action

The Parallel Game

- slower inclusion in expectation;
- rational only with compensating utility

It follows that:

- the two games are substitute methods of coordination
- one relies on public impersonal incentives, the other on private relational credibility
- both allow beneficial exchange under the threat of counterparty revision
- rational choice therefore reveals the marginal rate of substitution between them

EQUILIBRIUM LOGIC

The role of the mechanism is to enforce **directional coherence** across games.

Bid → Speed → Security

Without this coupling:

- a shift into the parallel game reveals nothing;
- one opaque utility bundle is preferred to another;
- there is no identifiable direction of substitution

With it:

- one entire class of explanations disappears;
- preferring the parallel game identifies an unambiguous preference
- counterparty credibility must be more valuable on the margin;
- or the game choice itself is irrational.

We have a complementary revealed-preference experiment:

- game choice identifies the relevant margin of substitution;
- bid choice reveals the preferred quantity of security along that margin;
- yet the mechanism never needs ask players to reveal valuations;
- as choosing the wrong game is costly in expectation.

The mechanism induces the rational and efficient provision of its baseline security function relative to an unobservable but strategically-relevant substitute (trust in counterparty credibility) that must be optimized against for most-efficient provision.

Efficient allocation under informational decentralization, contra Hurwicz (1972)

WHY REVELATION EQUIVALENCE FAILS

Revelation-equivalent mechanisms cannot preserve the directional coherence required to implement this equilibrium. Observe:

$$\boxed{\text{Bid} \longrightarrow \text{Speed} \longrightarrow \text{Security}}$$

$$F \rightarrow T \rightarrow E$$

Once we add an exogenous informational assumption (fixed type distribution, common priors, etc) our mapping from bids to security is no longer unique as we introduce an additional determinant of equilibrium X .

Our mechanism induces:

$$F \rightarrow (T, E)$$

The auxiliary assumption induces:

$$X \rightarrow E.$$

Game choice no longer identifies the direction of marginal substitution.

Conclusion.

- efficient allocation under informational decentralization is possible;
- non-revelation-equivalent mechanisms can preserve directional coherence;
- a reference mechanism is known that implements this equilibrium